SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA5117

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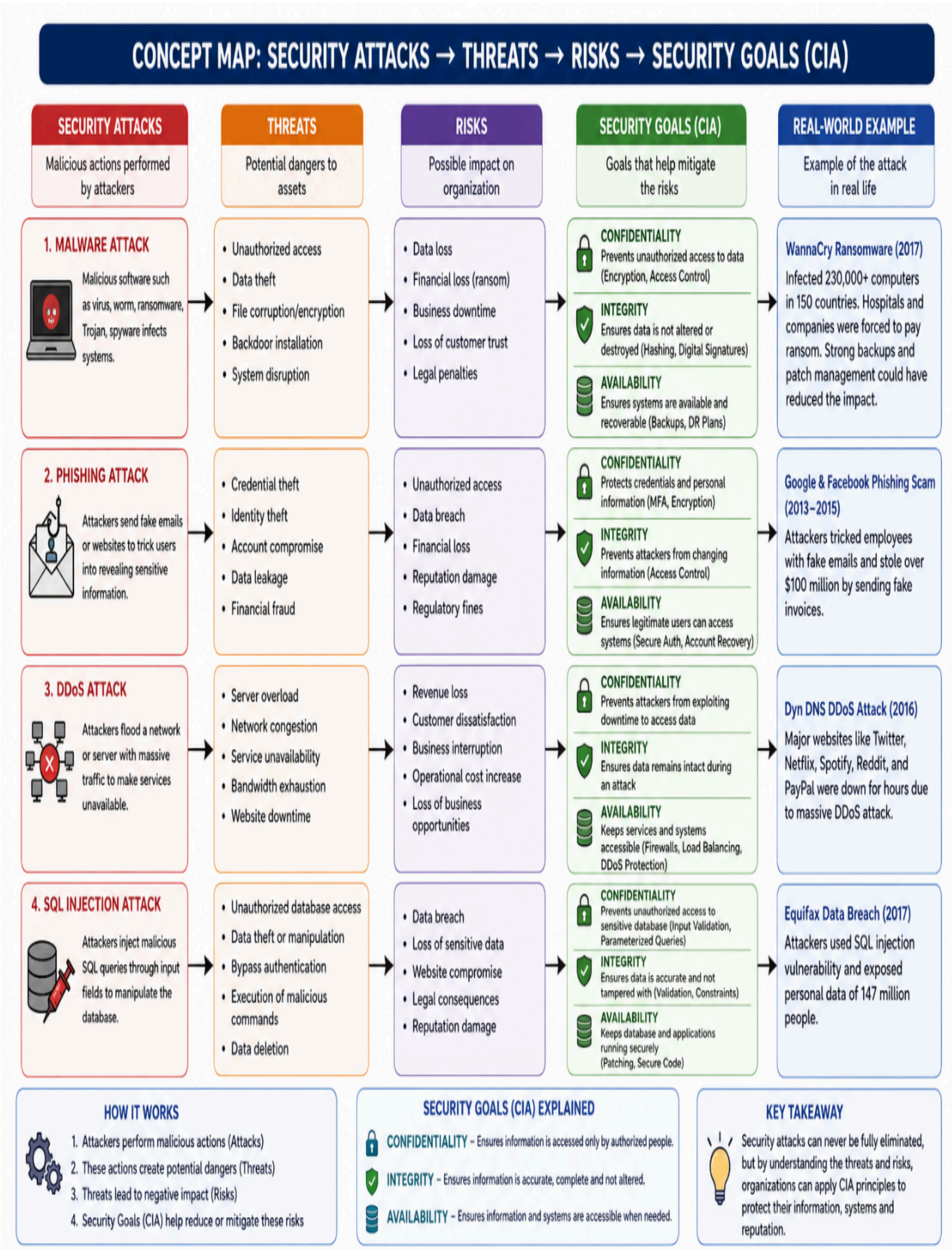
1.



CONCEPT MAP: SECURITY ATTACKS → THREATS → RISKS → SECURITY GOALS (CIA)
SECURITY ATTACKS
Malicious actions performed by attackers
THREATS
Potential dangers to assets
RISKS
Possible impact on organization
SECURITY GOALS (CIA)
Goals that help mitigate the risks
REAL-WORLD EXAMPLE
Example of the attack in real life
1. MALWARE ATTACK
Malicious software such as virus, worm, ransomware, Trojan, spyware infects systems.
• Unauthorized access
• Data theft
• File corruption/encryption
• Backdoor installation
• System disruption
• Data loss
• Financial loss (ransom)
• Business downtime
• Loss of customer trust
• Legal penalties
CONFIDENTIALITY
Prevents unauthorized access to data (Encryption, Access Control)
INTEGRITY
Ensures data is not altered or destroyed (Hashing, Digital Signatures)
AVAILABILITY
Ensures systems are available and recoverable (Backups, DR Plans)
WannaCry Ransomware (2017)
Infected 230,000+ computers in 150 countries. Hospitals and companies were forced to pay ransom. Strong backups and patch management could have reduced the impact.
2. PHISHING ATTACK
Attackers send fake emails or websites to trick users into revealing sensitive information.
• Credential theft
• Identity theft
• Account compromise
• Data leakage
• Financial fraud
• Unauthorized access
• Data breach
• Financial loss
• Reputation damage
• Regulatory fines
CONFIDENTIALITY
Protects credentials and personal information (MFA, Encryption)
INTEGRITY
Prevents attackers from changing information (Access Control)
AVAILABILITY
Ensures legitimate users can access systems (Secure Auth, Account Recovery)
Google & Facebook Phishing Scam (2013–2015)
Attackers tricked employees with fake emails and stole over $100 million by sending fake invoices.
3. DDoS ATTACK
Attackers flood a network or server with massive traffic to make services unavailable.
• Server overload
• Network congestion
• Service unavailability
• Bandwidth exhaustion
• Website downtime
• Revenue loss
• Customer dissatisfaction
• Business interruption
• Operational cost increase
• Loss of business opportunities
CONFIDENTIALITY
Prevents attackers from exploiting downtime to access data
INTEGRITY
Ensures data remains intact during an attack
AVAILABILITY
Keeps services and systems accessible (Firewalls, Load Balancing, DDoS Protection)
Dyn DNS DDoS Attack (2016)
Major websites like Twitter, Netflix, Spotify, Reddit, and PayPal were down for hours due to massive DDoS attack.
4. SQL INJECTION ATTACK
Attackers inject malicious SQL queries through input fields to manipulate the database.
• Unauthorized database access
• Data theft or manipulation
• Bypass authentication
• Execution of malicious commands
• Data deletion
• Data breach
• Loss of sensitive data
• Website compromise
• Legal consequences
• Reputation damage
CONFIDENTIALITY
Prevents unauthorized access to sensitive database (Input Validation, Parameterized Queries)
INTEGRITY
Ensures data is accurate and not tampered with (Validation, Constraints)
AVAILABILITY
Keeps database and applications running securely (Patching, Secure Code)
Equifax Data Breach (2017)
Attackers used SQL injection vulnerability and exposed personal data of 147 million people.
HOW IT WORKS
1. Attackers perform malicious actions (Attacks)
2. These actions create potential dangers (Threats)
3. Threats lead to negative impact (Risks)
4. Security Goals (CIA) help reduce or mitigate these risks
SECURITY GOALS (CIA) EXPLAINED
CONFIDENTIALITY – Ensures information is accessed only by authorized people.
INTEGRITY – Ensures information is accurate, complete and not altered.
AVAILABILITY – Ensures information and systems are accessible when needed.
KEY TAKEAWAY
Security attacks can never be fully eliminated, but by understanding the threats and risks, organizations can apply CIA principles to protect their information, systems and reputation.

Vulnerability Analysis

- **Substitution Ciphers**
    - **More vulnerable because statistical properties of language (like frequency of 'E' in English) remain visible.**
    - **Attackers can use frequency analysis to guess mappings.**
- **Transposition Ciphers**
    - **Slightly less vulnerable since they shuffle positions, making direct frequency analysis harder.**
    - **However, they can still be broken with pattern recognition and brute force.**

👉 **Overall, substitution is weaker because it directly exposes frequency patterns, while transposition at least hides them through rearrangement. But both are insecure by modern standards.**

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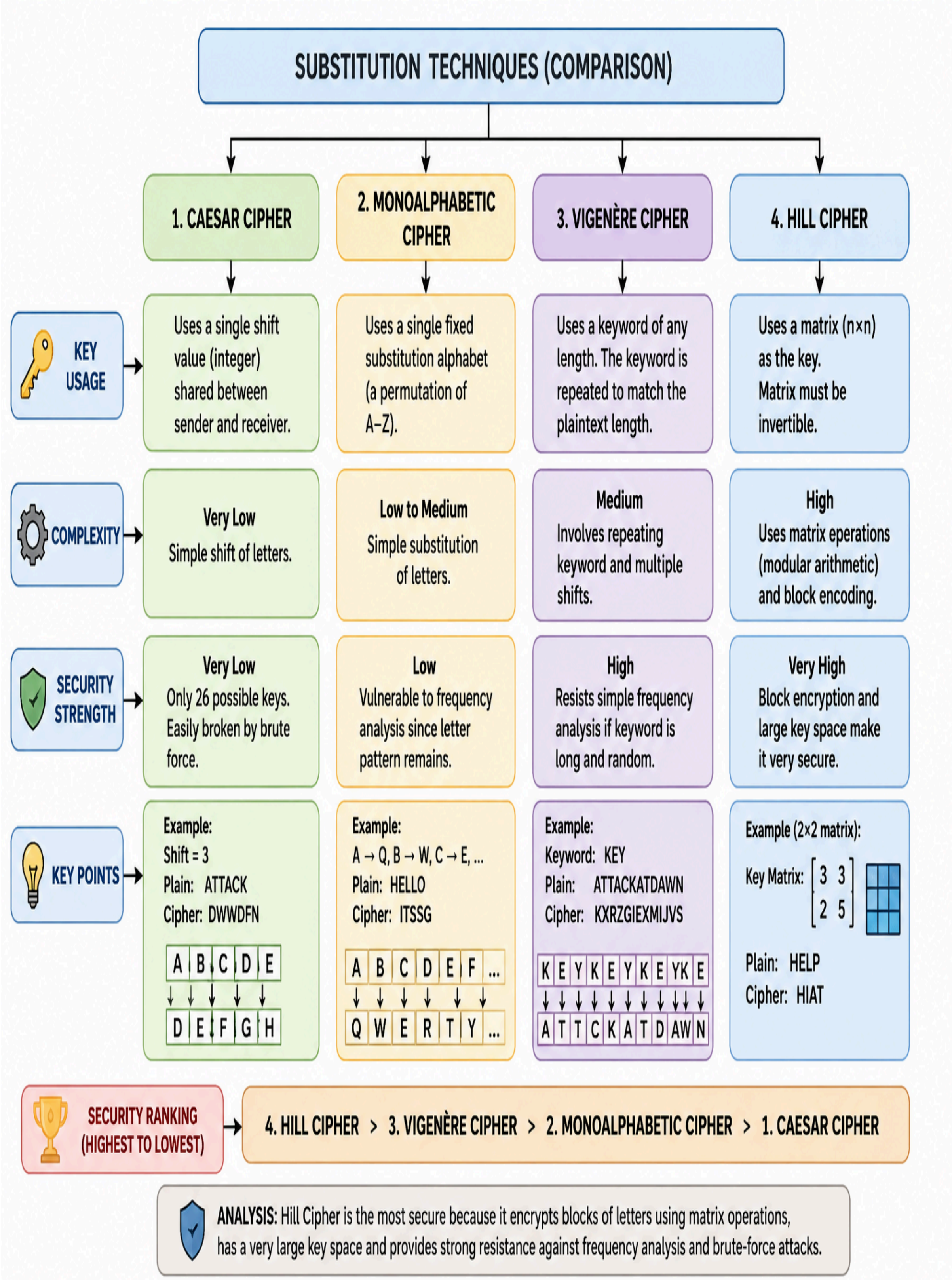

SUBSTITUTION TECHNIQUES (COMPARISON)
1. CAESAR CIPHER
2. MONOALPHABETIC CIPHER
3. VIGENÈRE CIPHER
4. HILL CIPHER
KEY USAGE
Uses a single shift value (integer) shared between sender and receiver.
Uses a single fixed substitution alphabet (a permutation of A–Z).
Uses a keyword of any length. The keyword is repeated to match the plaintext length.
Uses a matrix (n×n) as the key. Matrix must be invertible.
COMPLEXITY
Very Low
Simple shift of letters.
Low to Medium
Simple substitution of letters.
Medium
Involves repeating keyword and multiple shifts.
High
Uses matrix eperations (modular arithmetic) and block encoding.
SECURITY STRENGTH
Very Low
Only 26 possible keys. Easily broken by brute force.
Low
Vulnerable to frequency analysis since letter pattern remains.
High
Resists simple frequency analysis if keyword is long and random.
Very High
Block encryption and large key space make it very secure.
KEY POINTS
Example:
Shift = 3
Plain: ATTACK
Cipher: DWWDFN
A B C D E
D E F G H
Example:
A → Q, B → W, C → E, ...
Plain: HELLO
Cipher: ITSSG
A B C D E F ...
Q W E R T Y ...
Example:
Keyword: KEY
Plain: ATTACKATDAWN
Cipher: KXRZGIEXMIJVS
K E Y K E Y K E YK E
A T T C K A T D AW N
Example (2×2 matrix):
Key Matrix: 3 3 2 5
Plain: HELP
Cipher: HIAT
SECURITY RANKING (HIGHEST TO LOWEST)
4. HILL CIPHER > 3. VIGENÈRE CIPHER > 2. MONOALPHABETIC CIPHER > 1. CAESAR CIPHER
ANALYSIS: Hill Cipher is the most secure because it encrypts blocks of letters using matrix operations, has a very large key space and provides strong resistance against frequency analysis and brute-force attacks.

Analysis: Which Technique is Most Secure?

Among the four substitution techniques, the Hill Cipher is the most secure. It uses a matrix-based key to encrypt groups of letters instead of encrypting one letter at a time. This block encryption hides letter patterns effectively and makes traditional attacks such as frequency analysis much harder. It also has a much larger key space than the Caesar and Monoalphabetic ciphers.

The Vigenère Cipher is more secure than the Caesar and Monoalphabetic ciphers because it uses a repeating keyword, producing different shifts for different letters. However, if the keyword is short or repeated frequently, it can still be broken using cryptanalysis techniques.

The Monoalphabetic Cipher offers slightly better security than the Caesar Cipher because it uses a random substitution alphabet. However, it remains vulnerable to frequency analysis since each plaintext letter always maps to the same ciphertext letter.

The Caesar Cipher is the least secure because it uses only one shift value, resulting in just 26 possible keys. An attacker can easily recover the original message using brute-force or simple pattern analysis.

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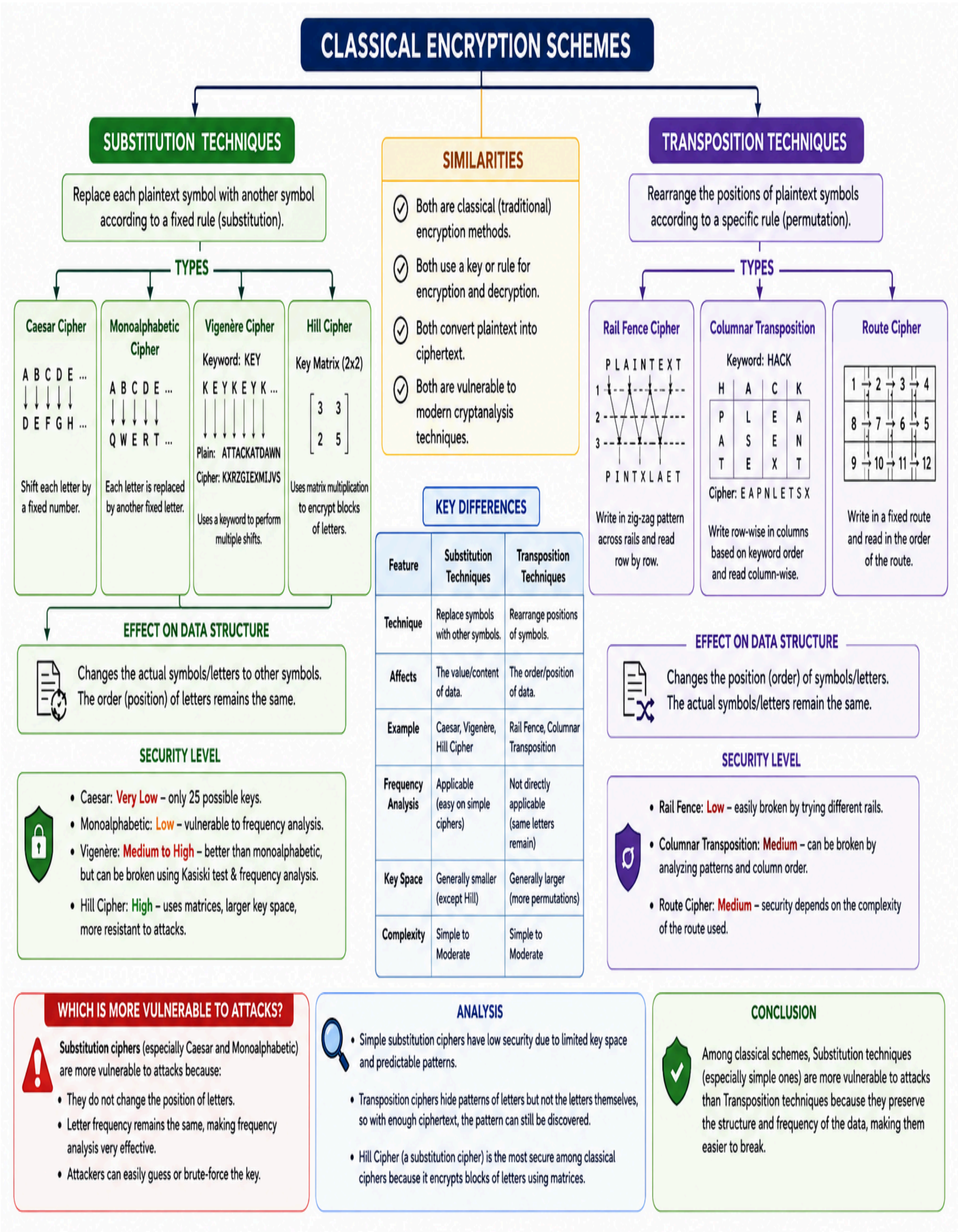
CLASSICAL ENCRYPTION SCHEMES
SUBSTITUTION TECHNIQUES
Replace each plaintext symbol with another symbol according to a fixed rule (substitution).
TYPES
Caesar Cipher
A B C D E ...
D E F G H ...
Shift each letter by a fixed number.
Monoalphabetic Cipher
A B C D E ...
Q W E R T ...
Each letter is replaced by another fixed letter.
Vigenère Cipher
Keyword: KEY
K E Y K E Y K ...
Plain: ATTACKATDAWN
Cipher: KXRZGIEXMIJVS
Uses a keyword to perform multiple shifts.
Hill Cipher
Key Matrix (2x2)
3 3
2 5
Uses matrix multiplication to encrypt blocks of letters.
SIMILARITIES
Both are classical (traditional) encryption methods.
Both use a key or rule for encryption and decryption.
Both convert plaintext into ciphertext.
Both are vulnerable to modern cryptanalysis techniques.
TRANSPOSITION TECHNIQUES
Rearrange the positions of plaintext symbols according to a specific rule (permutation).
TYPES
Rail Fence Cipher
P L A I N T E X T
1
2
3
P I N T X L A E T
Write in zig-zag pattern across rails and read row by row.
Columnar Transposition
Keyword: HACK
H A C K
P L E A
A S E N
T E X T
Cipher: E A P N L E T S X
Write row-wise in columns based on keyword order and read column-wise.
Route Cipher
1 → 2 → 3 → 4
8 → 7 → 6 → 5
9 → 10 → 11 → 12
Write in a fixed route and read in the order of the route.
KEY DIFFERENCES
Feature | Substitution Techniques | Transposition Techniques
Technique | Replace symbols with other symbols. | Rearrange positions of symbols.
Affects | The value/content of data. | The order/position of data.
Example | Caesar, Vigenère, Hill Cipher | Rail Fence, Columnar Transposition
Frequency Analysis | Applicable (easy on simple ciphers) | Not directly applicable (same letters remain)
Key Space | Generally smaller (except Hill) | Generally larger (more permutations)
Complexity | Simple to Moderate | Simple to Moderate
EFFECT ON DATA STRUCTURE
Changes the actual symbols/letters to other symbols. The order (position) of letters remains the same.
SECURITY LEVEL
Caesar: Very Low – only 25 possible keys.
Monoalphabetic: Low – vulnerable to frequency analysis.
Vigenère: Medium to High – better than monoalphabetic, but can be broken using Kasiski test & frequency analysis.
Hill Cipher: High – uses matrices, larger key space, more resistant to attacks.
EFFECT ON DATA STRUCTURE
Changes the position (order) of symbols/letters. The actual symbols/letters remain the same.
SECURITY LEVEL
Rail Fence: Low – easily broken by trying different rails.
Columnar Transposition: Medium – can be broken by analyzing patterns and column order.
Route Cipher: Medium – security depends on the complexity of the route used.
WHICH IS MORE VULNERABLE TO ATTACKS?
Substitution ciphers (especially Caesar and Monoalphabetic) are more vulnerable to attacks because:
They do not change the position of letters.
Letter frequency remains the same, making frequency analysis very effective.
Attackers can easily guess or brute-force the key.
ANALYSIS
Simple substitution ciphers have low security due to limited key space and predictable patterns.
Transposition ciphers hide patterns of letters but not the letters themselves, so with enough ciphertext, the pattern can still be discovered.
Hill Cipher (a substitution cipher) is the most secure among classical ciphers because it encrypts blocks of letters using matrices.
CONCLUSION
Among classical schemes, Substitution techniques (especially simple ones) are more vulnerable to attacks than Transposition techniques because they preserve the structure and frequency of the data, making them easier to break.

Vulnerability Analysis

- **Substitution Ciphers**
  - **More vulnerable because statistical properties of language (like frequency of 'E' in English) remain visible.**
  - **Attackers can use frequency analysis to guess mappings.**
- **Transposition Ciphers**
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👉 **Overall, substitution is weaker because it directly exposes frequency patterns, while transposition at least hides them through rearrangement. But both are insecure by modern standards.**

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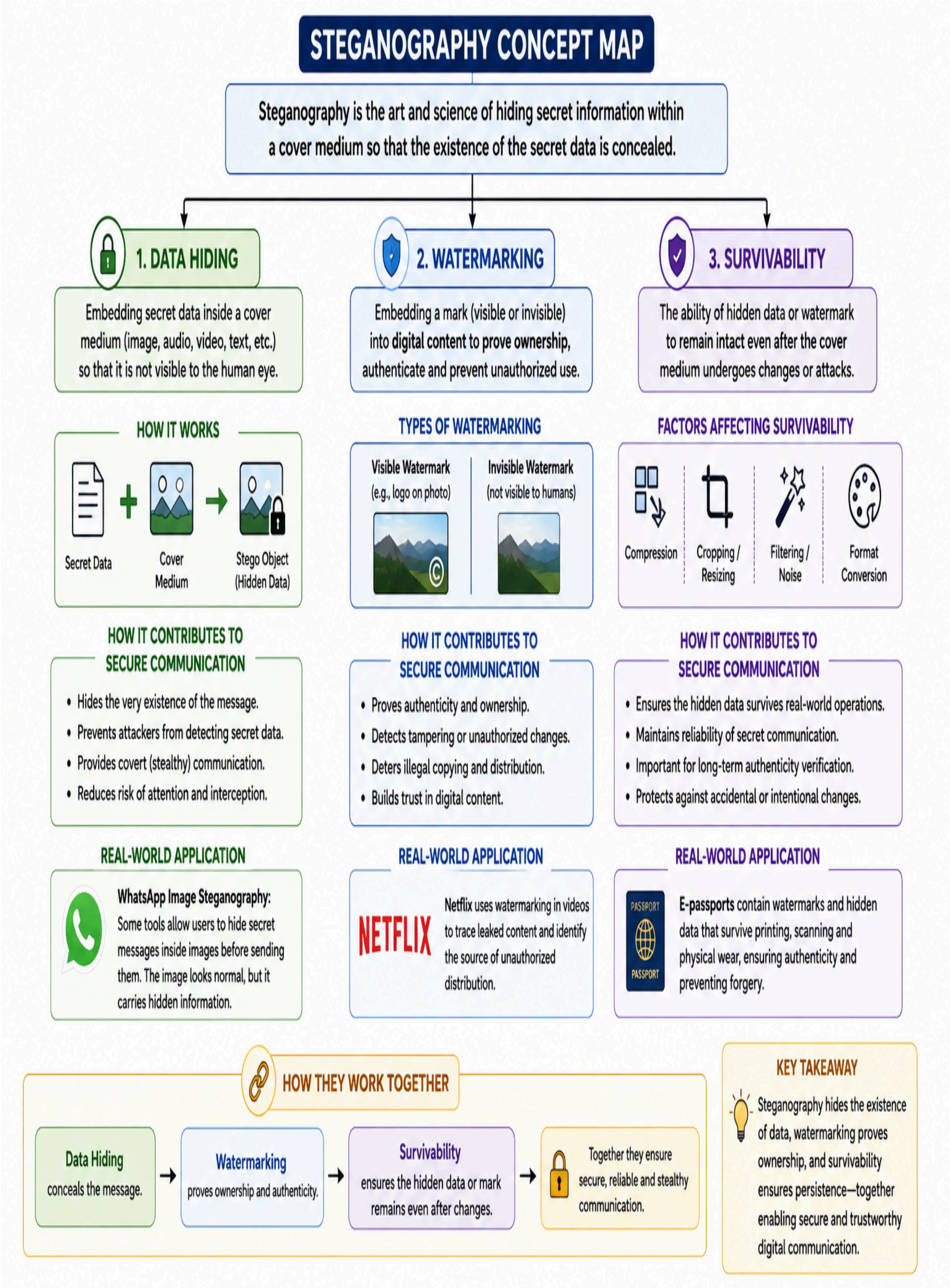

STEGANOGRAPHY CONCEPT MAP
Steganography is the art and science of hiding secret information within a cover medium so that the existence of the secret data is concealed.
1. DATA HIDING
Embedding secret data inside a cover medium (image, audio, video, text, etc.) so that it is not visible to the human eye.
2. WATERMARKING
Embedding a mark (visible or invisible) into digital content to prove ownership, authenticate and prevent unauthorized use.
3. SURVIVABILITY
The ability of hidden data or watermark to remain intact even after the cover medium undergoes changes or attacks.
HOW IT WORKS
Secret Data
Cover Medium
Stego Object (Hidden Data)
TYPES OF WATERMARKING
Visible Watermark (e.g., logo on photo)
Invisible Watermark (not visible to humans)
FACTORS AFFECTING SURVIVABILITY
Compression
Cropping / Resizing
Filtering / Noise
Format Conversion
HOW IT CONTRIBUTES TO SECURE COMMUNICATION
• Hides the very existence of the message.
• Prevents attackers from detecting secret data.
• Provides covert (stealthy) communication.
• Reduces risk of attention and interception.
HOW IT CONTRIBUTES TO SECURE COMMUNICATION
• Proves authenticity and ownership.
• Detects tampering or unauthorized changes.
• Deters illegal copying and distribution.
• Builds trust in digital content.
HOW IT CONTRIBUTES TO SECURE COMMUNICATION
• Ensures the hidden data survives real-world operations.
• Maintains reliability of secret communication.
• Important for long-term authenticity verification.
• Protects against accidental or intentional changes.
REAL-WORLD APPLICATION
WhatsApp Image Steganography: Some tools allow users to hide secret messages inside images before sending them. The image looks normal, but it carries hidden information.
REAL-WORLD APPLICATION
NETFLIX
Netflix uses watermarking in videos to trace leaked content and identify the source of unauthorized distribution.
REAL-WORLD APPLICATION
PASSPORT
PASSPORT
E-passports contain watermarks and hidden data that survive printing, scanning and physical wear, ensuring authenticity and preventing forgery.
HOW THEY WORK TOGETHER
Data Hiding
conceals the message.
Watermarking
proves ownership and authenticity.
Survivability
ensures the hidden data or mark remains even after changes.
Together they ensure secure, reliable and stealthy communication.
KEY TAKEAWAY
Steganography hides the existence of data, watermarking proves ownership, and survivability ensures persistence—together enabling secure and trustworthy digital communication.

Vulnerability Analysis

- **Data Hiding Vulnerabilities**
  - **Susceptible to steganalysis (statistical detection of hidden patterns).**
  - **Weak if cover media is small or altered (e.g., resizing images).**
- **Watermarking Vulnerabilities**
  - **Can be removed or tampered with using signal processing attacks (cropping, filtering).**
  - **Visible watermarks may reduce usability; invisible ones may be erased.**
- **Survivability Vulnerabilities**
  - **Strong compression or format conversion may destroy hidden data.**
  - **Attackers can deliberately apply transformations to erase embedded content.**

👉 **Overall: Data hiding is most vulnerable to detection attacks, watermarking is vulnerable to removal attacks, and survivability is vulnerable to transformation attacks.**

5.

NUMBER THEORY CONCEPT MAP IN CRYPTOGRAPHY

These concepts are interconnected and form the mathematical foundation for public key cryptographic systems like RSA.

CONNECTIONS

- → Directly used / Depends on
- - - → Derived from / Special case
- ····→ Supports / Enables

Uses EEA to find inverse

Uses $\phi(n)$ in the theorem

Special case when n is prime

1. MODULAR INVERSE

For integers a and m, a^{-1} (mod m) exists if $\gcd(a, m) = 1$, such that: $a \cdot a^{-1} \equiv 1 \pmod{m}$

2. EXTENDED EUCLID ALGORITHM

Finds $\gcd(a, m)$ and integers x, y such that: $a\,x + m\,y = \gcd(a, m)$

If $\gcd(a, m) = 1$, then x is the modular inverse of a mod m.

3. EULER'S PHI FUNCTION φ(n)

Counts the number of integers less than or equal to n that are coprime to n.

4. EULER'S THEOREM

If $\gcd(a, n) = 1$, then: $a^{\phi(n)} \equiv 1 \pmod{n}$

5. FERMAT'S LITTLE THEOREM (FLT)

If p is prime and a not divisible by p, then: $a^{p-1} \equiv 1 \pmod{p}$

EEA is the method to compute inverse

$\phi(n)$ is used in Euler's theorem

FLT is a special case of Euler's theorem when $n = p$ (prime), since $\phi(p) = p - 1$

ROLE IN ENCRYPTION / DECRYPTION (RSA EXAMPLE)

Key Generation

Choose primes p and q. Compute $n = p \times q$ and $\phi(n) = (p-1)(q-1)$ using Euler's Phi Function.

Choose Public and Private Keys

Choose e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$. Use Extended Euclid Algorithm to find d (the modular inverse of e mod $\phi(n)$), i.e., $e \cdot d \equiv 1 \pmod{\phi(n)}$

Encryption

Public key (n, e) is used. For message M ($< n$):

$C \equiv M^e \pmod{n}$

(Uses Modular exponentiation and properties from Euler's Theorem)

Decryption

Private key (n, d) is used. For ciphertext C:

$M \equiv C^d \pmod{n}$

Because $e \cdot d \equiv 1 \pmod{\phi(n)}$, by Euler's Theorem:

$C^d \equiv (M^e)^d \equiv M^{ed} \equiv M \pmod{n}$

Correctness Guarantee

Euler's Theorem ensures encryption and decryption are inverse operations. Fermat's Little Theorem is used when n is prime (e.g., in some cryptosystems like Diffie-Hellman with prime modulus).

HOW THEY WORK TOGETHER (MATHEMATICAL FLOW)

EEA — Finds gcd and modular inverse → **Modular Inverse** — Needed to compute private key (d) → **Euler's Phi $\phi(n)$** — Counts coprimes and used in Euler's Theorem → **Euler's Theorem** — Enables correct encryption/ decryption → **Fermat's Little Theorem** — Special case when n is prime

All ensure the existence of d such that $e \cdot d \equiv 1 \pmod{\phi(n)}$

WHICH CONCEPT IS MOST CRITICAL FOR PUBLIC KEY CRYPTOGRAPHY?

Euler's Theorem (and Euler's Phi Function) is the most critical.

Why?

- RSA and many other public key systems rely on the fact that $a^{\phi(n)} \equiv 1 \pmod{n}$ for numbers coprime with n.
- This property guarantees that encryption and decryption are inverse operations.
- Without Euler's Theorem (and $\phi(n)$), we cannot mathematically prove the correctness of RSA.
- Modular Inverse and Extended Euclid Algorithm are essential tools, but Euler's Theorem provides the theoretical foundation that makes the entire system work.

RSA Security

SUMMARY: Extended Euclid Algorithm helps compute Modular Inverse. Euler's Phi Function counts coprime integers. Euler's Theorem uses $\phi(n)$ to create a mathematical relationship for secure encryption and decryption. Fermat's Little Theorem is a special case of Euler's Theorem for prime moduli.

Vulnerability Analysis

- **Modular Inverse → Vulnerable if modulus is not chosen properly (non-coprime values break security).**
- **Extended Euclid Algorithm → Computationally safe, but if gcd ≠ 1, inverse doesn't exist → weak keys.**
- **Fermat's Little Theorem → Vulnerable to Carmichael numbers (false positives in primality testing).**
- **Euler's Phi Function → Vulnerable if factorization of n is easy (φ(n) reveals private key).**
- **Euler's Theorem → Secure only if modulus n is hard to factor; otherwise RSA collapses.**

👉 **Most Critical Concept: Euler's Theorem**

- **Justification: It is the mathematical backbone of RSA. Without Euler's Theorem, the encryption/decryption cycle (c^d ≡ m mod n) would not hold. While modular inverses and φ(n) are essential tools, Euler's Theorem provides the fundamental guarantee of correctness and security in public key cryptography.**